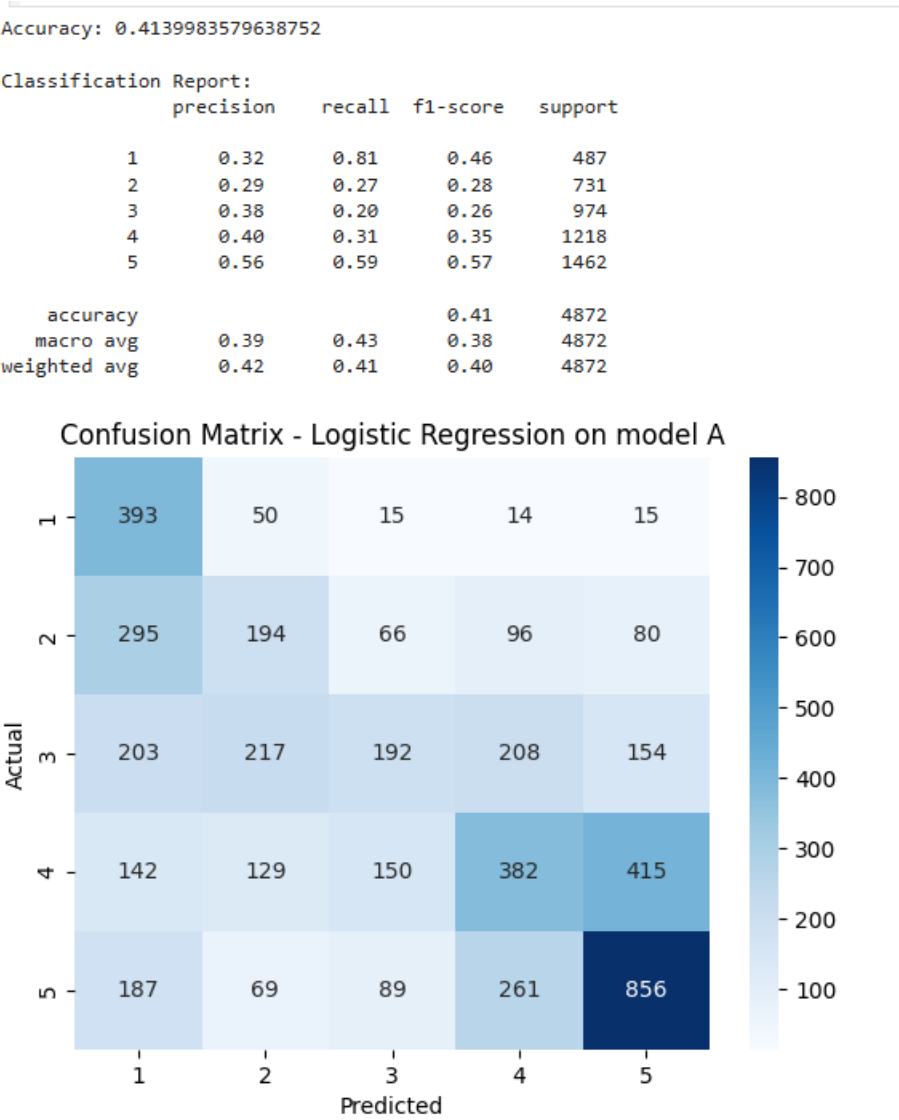


CROSS TEST REPORT

SUMMARY

In this project, two models were trained and evaluated. Model A achieved an accuracy of 41%, whereas Model B achieved 35% accuracy. Although both models showed relatively low performance, Model A performed better than Model B .

Model A tested using imbalanced dataset:

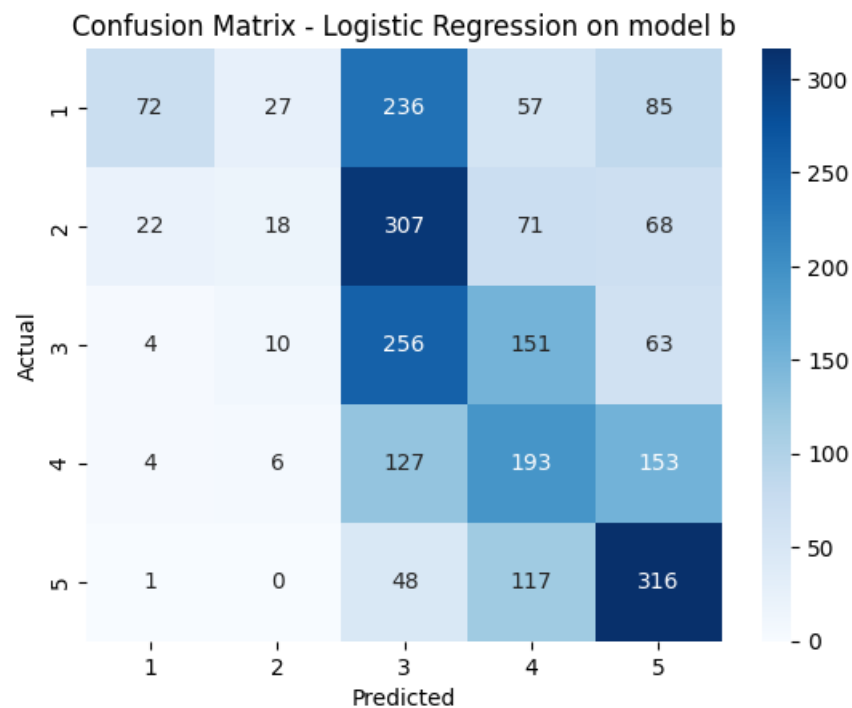


Model B tested using the balanced dataset:

Accuracy: 0.35447761194029853

Classification Report:

	precision	recall	f1-score	support
1	0.70	0.15	0.25	477
2	0.30	0.04	0.07	486
3	0.26	0.53	0.35	484
4	0.33	0.40	0.36	483
5	0.46	0.66	0.54	482
accuracy			0.35	2412
macro avg	0.41	0.35	0.31	2412
weighted avg	0.41	0.35	0.31	2412



OBSERVATIONS

- When trained on one dataset and tested on a different dataset, the performance of both models dropped sharply.
- The drop indicates poor generalization across domains.
- Model B, even though it scored 57% before cross-test, still struggled heavily on unfamiliar data.

- The model accuracy dropped because each dataset uses different vocabulary, style, and context, so the model learned patterns specific to the training domain.
- This shows the need for combined or merged datasets for better cross-domain performance.

RECOMMENDATION: choosing the model

Based on initial evaluation, Model A should be deployed because it has better accuracy (41%) compared to Model B (35%) on its own test set. Even though both models struggle in cross testing. Between the two, Model A is recommended for deployment because it performed better overall with a higher accuracy (41%) and slightly better consistency in predictions.

Although the performance is still not ideal, Model A is the best choice among the options tested and can serve as a baseline model. It can be deployed with the intention of further improvement through hyperparameter tuning and balanced training data in future work.